AP PRECALCULUS

Unit 1 • Worksheet 1.13-1.14A

Worksheet 1.13-1.14A • Function Model Construction

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Data with equally spaced inputs have first differences $4.9,\ 5.1,\ 5.0,\ 4.9$. Which model fits best?

(A) linear
(B) quadratic
(C) cubic
(D) rational

Question 2

NO CALCULATOR

A company's average cost per item is $A(n) = \frac{500+4n}{n}$. What does $\lim_{n\to\infty} A(n)$ tell you?

(A) The fixed cost is \$500.
(B) The average cost becomes 0.
(C) The average cost per item approaches \$4.
(D) The cost per item grows without bound.

Question 3

NO CALCULATOR

A rectangle has perimeter 40 cm and width w. Which gives its area?

(A) $w(40 - w)$
(B) $w(20 - w)$
(C) $2w + 40$
(D) $w^2 - 20$

Question 4

CALCULATOR

The height of a model rocket is recorded: $(0, 1.2)$, $(1, 6.1)$, $(2, 9.0)$, $(3, 9.9)$, $(4, 8.6)$ (t in s, h in m). Using quadratic regression, what is the predicted height at $t = 5$?

(A) 4.2 m
(B) 5.3 m
(C) 6.4 m
(D) 7.3 m

Question 5 (Free response) CALCULATOR

A cylindrical can holds 500 cm^3. Let r be the radius and h the height (cm).

(a) Express h in terms of r, and show that the surface area is $S(r) = 2\pi r^2 + \frac{1000}{r}$.
(b) Find the radius that minimizes S, and the minimum surface area.
(c) Describe S as $r \to 0^+$ and $r \to \infty$, and explain in context.

Question 6 (Free response) NO CALCULATOR

A phone plan costs \$20 per month including 100 texts, plus \$0.10 per extra text.

(a) Write a piecewise model $C(n)$ and find $C(250)$.
(b) For $n > 100$, find $\lim_{n\to\infty} \frac{C(n)}{n}$ and interpret it.